

Chemistry 12

Unit 1. Organic Chemistry

Overall Expectations

- ❖ Assess the social and environmental impact of organic compounds used in everyday life, and propose a course of action to reduce the use of compounds that are harmful to human health and the environment.
- ❖ Investigate organic compounds and organic chemical reactions, and use various methods to represent the compounds.
- ❖ Demonstrate an understanding of the structure, properties, and chemical behaviour of compounds with each class of organic compounds.

BIG IDEAS

- ❖ Organic compounds have predictable chemical and physical properties determined by their respective structures
- ❖ Organic chemical reactions and their applications have significant implications for society, human, health, and the environment.

Unit 2. Structure and properties of matter

Overall Expectations

- ❖ Assess the benefits to society and environmental impact of products and technologies that apply principles related to the structure and properties of matter.
- ❖ Investigate the molecule shapes and physical properties of various types of matter
- ❖ **Demonstrate** an understanding of atomic structure and chemical bonding, and how they relate to physical properties of ionic, molecular, covalent network, and metallic substances.

BIG IDEAS

- ❖ The nature of attractive forces that exists between particles in a substance determine the properties of that substance.
- ❖ *Technological devices that are based on the principles of atomic and molecular structure can have social benefits and costs.*

Unit 3. Energy changes and rates of reactions

Overall Expectations

- ❖ Analyze technological and chemical processes that are based on energy changes, and their efficiency and their effects on their environment.
- ❖ Investigate and analyze energy changes and rates of reaction in physical and chemical process. And solve related problems.
- ❖ Demonstrate an understanding of energy changes and rates of reaction.

BIG IDEAS

- ❖ Energy changes and rate of chemical reactions can be described quantitatively.
- ❖ The efficiency of chemical reactions can be improved by applying optimal conditions.
- ❖ Technologies that transform energy can have societal and environmental costs and benefits.

Unit 4. Chemical systems and equilibrium

Overall Expectations

- ❖ Analyze chemical equilibrium process, and assess their impact on biological, biochemical, and technological systems

- ❖ Investigate the qualitative and quantitative nature of chemical systems at equilibrium, and solve related problems.
- ❖ Demonstrate an understanding of the concept of dynamic equilibrium and the variables that cause shifts in the equilibrium of chemical systems.

BIG IDEAS

- ❖ Chemical systems are dynamic and respond to changing conditions in predictable ways.
- ❖ Applications of chemical systems at equilibrium have significant implications for nature and Industry.

Unit 5. Electrochemistry

Overall Expectations

- ❖ Analyze technologies and process relating to electrochemistry, and their implications relating to electrochemistry, and their implications for society, health and safety, the environment.
- ❖ Investigate oxidation-reduction reactions using a galvanic cell, and analyze electrochemistry reactions in qualitative and quantitative terms
- ❖ Demonstrate an understanding of the principles of oxidation-reduction reaction and the many practical applications of electrochemistry.

BIG IDEAS

- ❖ Oxidation and reduction are paired chemical reactions in which electrons are transferred from one substance to another in a predictable way.
- ❖ The control and application of oxidation and reduction reactions have significant implications for industry, health, and safety, and the environment.